



**EE 350 – Electronics II**

**Spring 2025/26**

**LAB 4**

**SUCCESSIVE APPROXIMATION REGISTER ADC**

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# 1 Overview

In this laboratory session, a Successive Approximation Register (SAR) Analog-to-Digital Converter (ADC) was physically implemented and tested. An Arduino microcontroller served as the SAR controller, producing a software-based PWM signal that was converted into a continuous DC reference voltage through an RC low-pass filter ( $R = 47 \text{ k}\Omega$ ,  $C = 10 \text{ }\mu\text{F}$ ). An MCP6022 Op-amp, configured as a comparator, evaluated whether the generated estimate was higher or lower than the unknown input voltage ( $V_{in}$ ). The primary objective was to observe the step-by-step binary search approximation process and evaluate the effect of bit-depth resolution (4-bit vs. 10-bit) on conversion accuracy.

## 2 Hardware Implementation & Verification

### 2.1 Initial Setup

The power supply was set to approximately 2.7 V. Verification with a digital multimeter yielded an exact reference measurement of  $V_{in} = 2.752 \text{ V}$ . This measured value serves as the ground truth for calculating the system's Total Unadjusted Error (TUE) and assessing the accuracy of both conversion modes.

### 2.2 4-bit SAR ADC Conversion

The 4-bit SAR ADC conversion sequence was executed. At each of the 4 conversion steps, the theoretical expected voltage (PWM target) and the actual measured voltage across the capacitor were recorded. Table 1 presents these results. A fifth row shows the final settled output after the algorithm completes.

| Table 1: 4-bit SAR ADC Conversion Steps ( $V_{in} = 2.752 \text{ V}$ ) |            |                |                        |
|------------------------------------------------------------------------|------------|----------------|------------------------|
| Step                                                                   | Bit Tested | PWM Target (V) | Multimeter Reading (V) |
| 1                                                                      | Bit 3      | 2.500          | 2.538                  |
| 2                                                                      | Bit 2      | 3.750          | 3.798                  |
| 3                                                                      | Bit 1      | 3.125          | 3.116                  |
| 4                                                                      | Bit 0      | 2.812          | 2.852                  |
| Final                                                                  | 1000       | 2.500          | 2.539                  |

Since  $V_{in} > 2.500 \text{ V}$ , the MSB (Bit 3) is kept. However, the subsequent test values (3.750, 3.125, 2.812 V) all exceed  $V_{in}$ , so Bits 2, 1, and 0 are cleared. The final register value is 1000 (decimal 8), yielding a theoretical estimate of  $V_{est} = 8 \times 5.0/16 = 2.500 \text{ V}$ .

### 2.3 10-bit SAR ADC Conversion

To observe the impact of increased resolution, the conversion was repeated in 10-bit mode. Table 2 records all 10 steps, showing the convergence from the initial half-scale test (2.500 V) toward the 2.752 V reference.

Table 2: 10-bit SAR ADC Conversion Steps ( $V_{in} = 2.752\text{ V}$ )

| Step | Bit Tested | PWM Target (V) | Multimeter Reading (V) |
|------|------------|----------------|------------------------|
| 1    | Bit 9      | 2.500          | 2.538                  |
| 2    | Bit 8      | 3.750          | 3.792                  |
| 3    | Bit 7      | 3.125          | 3.169                  |
| 4    | Bit 6      | 2.812          | 2.852                  |
| 5    | Bit 5      | 2.656          | 2.694                  |
| 6    | Bit 4      | 2.734          | 2.775                  |
| 7    | Bit 3      | 2.695          | 2.734                  |
| 8    | Bit 2      | 2.715          | 2.752                  |
| 9    | Bit 1      | 2.725          | 2.761                  |
| 10   | Bit 0      | 2.720          | 2.762                  |

The 10-bit algorithm progressively narrowed the search range, reaching a final measured capacitor voltage of  $2.762\text{ V}$ . Notably, at Step 8 the measured voltage exactly matched the multimeter reference of  $2.752\text{ V}$ , demonstrating the high precision achievable with increased bit depth.

### 3 Data Visualization

The successive approximation algorithm exhibits a characteristic non-monotonic “zig-zag” behavior, as it constantly overshoots and undershoots the target voltage to refine its search boundaries. The following staircase plots illustrate the convergence of the estimated voltage toward the true input voltage at each step.

#### 3.1 4-bit SAR ADC Staircase

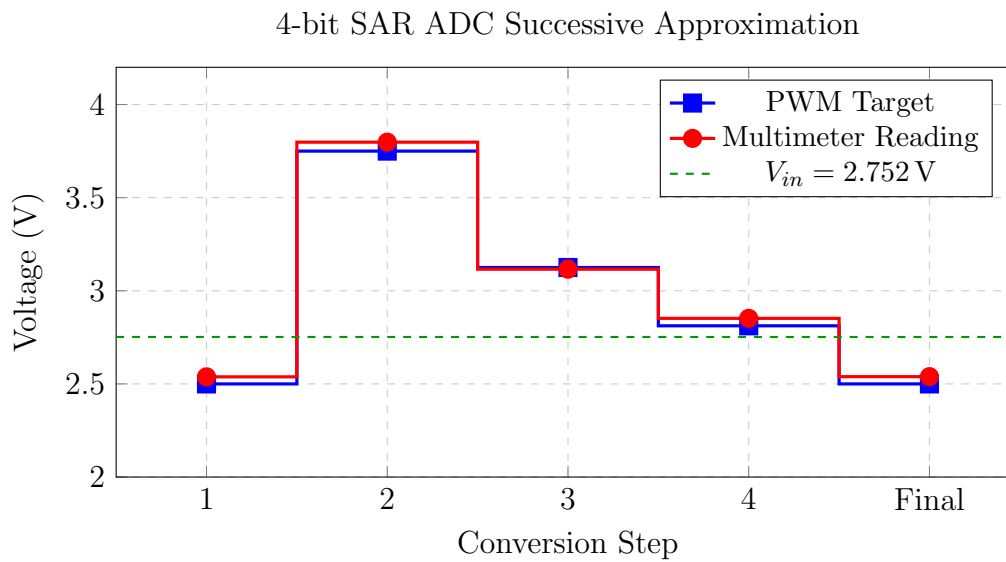


Figure 1: 4-bit SAR ADC Staircase Progression

#### 3.2 10-bit SAR ADC Staircase

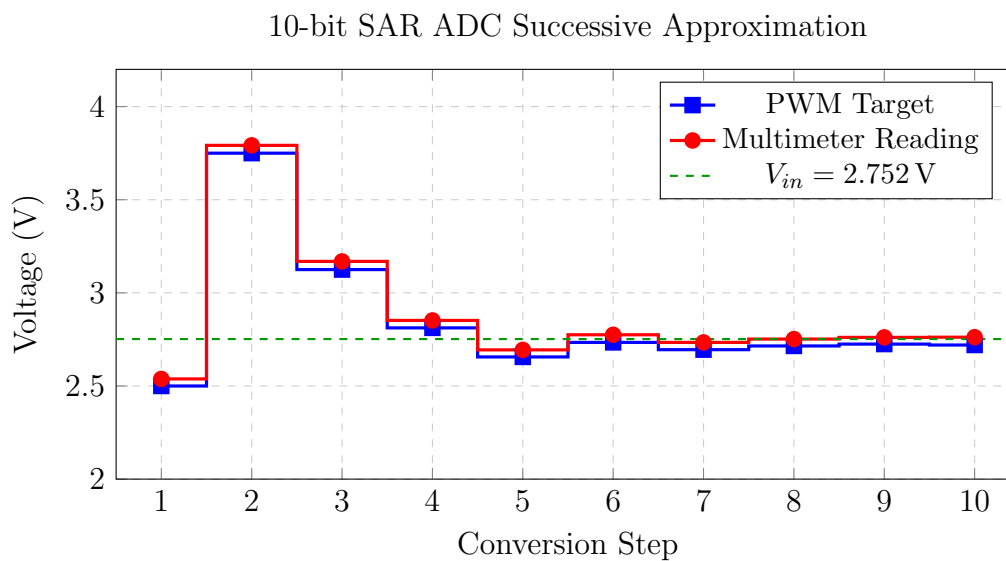


Figure 2: 10-bit SAR ADC Staircase Progression

## 4 Experimental Analysis

### 4.1 Duty Cycle & Mean Voltage Verification

At the conclusion of the 10-bit conversion, the PWM signal was captured via oscilloscope to verify the relationship between the digital logic and the analog output. Figure 3 shows the oscilloscope capture.

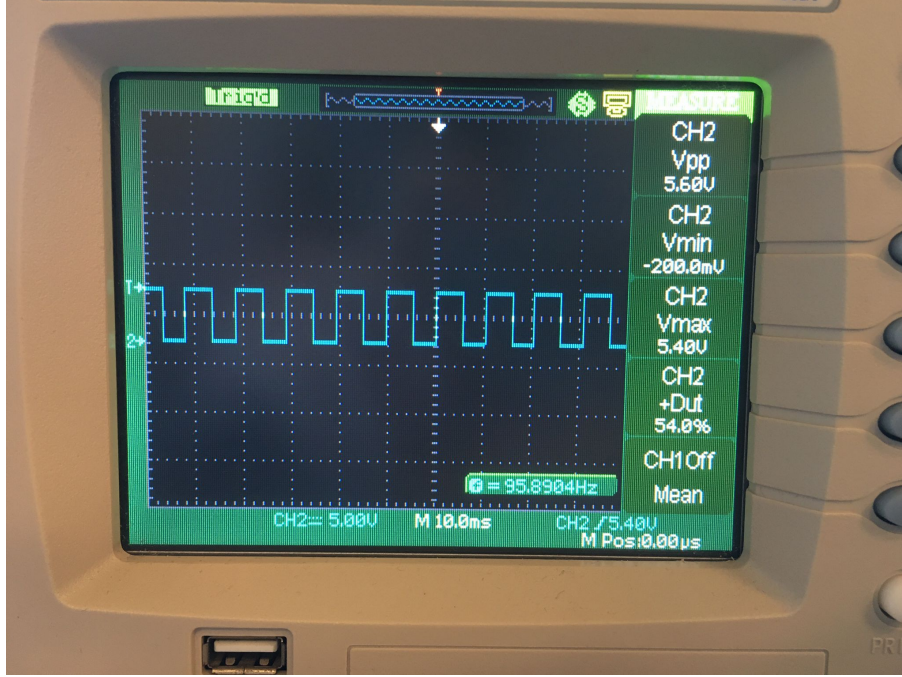


Figure 3: Oscilloscope capture of the final-step PWM signal showing a 54.0% duty cycle.

- **Measured Duty Cycle ( $D$ ):** 54%
- **Supply Voltage ( $V_{supply}$ ):** 5.00 V
- **Calculated Mean Voltage:**

$$V_{mean} = D \times V_{supply} = 0.54 \times 5.0 \text{ V} = 2.70 \text{ V}$$

The proximity of this calculated result to the 2.752 V input confirms that the SAR controller effectively modulated the PWM to balance the inputs of the MCP6022 comparator. The slight difference between the calculated mean voltage (2.70 V) and the actual capacitor voltage (2.762 V) is attributed to the inherent ripple of the RC filter and the finite resolution of the oscilloscope’s duty cycle measurement.

### 4.2 Accuracy Evaluation

Table 3: Accuracy comparison of 4-bit and 10-bit SAR ADC results.

| Mode   | $V_{in}$ (V) | Final Estimate (V) | Absolute Error (V) |
|--------|--------------|--------------------|--------------------|
| 4-bit  | 2.752        | 2.539              | 0.213              |
| 10-bit | 2.752        | 2.762              | 0.010              |

The 10-bit ADC reduces the absolute error from 213 mV to just 10 mV—a factor of  $\sim 21\times$  improvement. This clearly demonstrates that increasing the bit depth enables the SAR algorithm to converge much more closely to the true input voltage.

### 4.3 Total Unadjusted Error (TUE)

The TUE is calculated as the absolute difference between the true input voltage and the final ADC estimate. The theoretical quantization step size (1 LSB) and maximum quantization error for each resolution are:

$$\begin{aligned} \text{LSB}_4 &= \frac{5.0}{2^4} = 312.5 \text{ mV}, & Q_{e,max} &= \frac{\text{LSB}}{2} = 156.25 \text{ mV} \\ \text{LSB}_{10} &= \frac{5.0}{2^{10}} = 4.883 \text{ mV}, & Q_{e,max} &= \frac{\text{LSB}}{2} = 2.441 \text{ mV} \end{aligned}$$

Table 4: Total Unadjusted Error comparison.

| Mode   | TUE (mV) | Theoretical $Q_{e,max}$ (mV) | TUE / LSB |
|--------|----------|------------------------------|-----------|
| 4-bit  | 213.0    | 156.25                       | 0.68      |
| 10-bit | 10.0     | 2.441                        | 2.05      |

For the 4-bit ADC, the TUE of 213 mV falls within 1 LSB (312.5 mV), which is consistent with the theoretical quantization limit. The error is primarily dominated by the coarse resolution of the 4-bit system. For the 10-bit ADC, the TUE of 10 mV exceeds the theoretical quantization floor (2.441 mV) by approximately 2 LSBs, indicating that analog error sources—primarily the RC filter settling time ( $\tau = 47 \text{ k}\Omega \times 10 \text{ }\mu\text{F} = 0.47 \text{ s}$ ) and the comparator’s input offset voltage—contribute to the residual error at this resolution.

## 5 Conclusion

The laboratory experiment successfully demonstrated the implementation of a SAR ADC using an Arduino-based PWM-to-DC conversion technique. The convergence data and duty cycle measurements confirm that the binary search algorithm functions correctly, narrowing the error margin with each successive bit.

The 4-bit conversion yielded a final estimate of 2.539 V with an absolute error of 213 mV, while the 10-bit conversion reached 2.762 V with an error of only 10 mV. The staircase progression plots clearly illustrated the non-monotonic, zig-zag nature of the SAR algorithm, and the 10-bit zoomed view showed how the estimate oscillates around the true input with diminishing amplitude.

Although the 10-bit resolution provides a theoretical quantization floor of 4.88 mV, practical analog imperfections such as RC filter settling and comparator sensitivity set the final precision limit at 10 mV in this implementation. This demonstrates that while increasing bit depth significantly improves accuracy, the analog front-end must also be optimized to fully exploit the available digital resolution.